Aniket Balaji Parnale

Parbhani, Maharashtra, India
+91 7385020441 parnaleaniket@gmail.com linkedin.com/in/aniket-parnale



Education

Indian Institute of Space Science & Technology, Thiruvananthapuram **2025 – 2027**
M.Tech in Manufacturing Technology (SC25M141) *Thiruvananthapuram, Kerala*

Government College of Engineering, Chandrapur **2020 – 2024**
B.Tech in Mechanical Engineering (NBA Accredited) — CGPA: 7.64 *Chandrapur, Maharashtra*

Science Junior College, Parbhani **2008 – 2020**
Higher Secondary Certificate (HSC) — 59.08% *Parbhani, Maharashtra*

Nutan Vidyamandir Boy's And Girl's High School, Parbhani **2006 – 2018**
Secondary School Certificate (SSC) — 77.80% *Parbhani, Maharashtra*

Experience

Indian Institute of Space Science & Technology, Thiruvananthapuram **Summer 2026**
Research Intern – Friction Stir Welding (Dept. of Aerospace Engineering) *Thiruvananthapuram, Kerala*

- Carried out friction stir welding (FSW) trials on 6 mm AA1100 aluminium plates using a single-head vertical FSW machine (30 kN thrust, 0–3000 rpm, 0–1500 mm/min), under Dr. P. Chakravarthy
- Iteratively tuned spindle speed, traverse speed, plunge depth and fixture alignment to achieve a continuous, well-consolidated weld; diagnosed failed trials (low-RPM instability and flash, insufficient pin penetration) by modelling the pin as a cantilever under bending and torsion
- Tensile-tested four welded specimens per ASTM E8: UTS 87.31–92.85 MPa (mean 89.73 MPa); full-length weld gave the highest UTS (92.85 MPa) and elongation
- Performed Vickers microhardness mapping per ASTM E384 (HV0.1, 6 indents): 32.1–35.5 HV, mean 33.77 HV, SD 1.12 HV, $C_{pk} = 4.83$; prepared and examined metallographic samples (sectioning, mounting, polishing, etching, optical microscopy) in the IIST metallography lab
- Authored internship report and delivered seminar; reviewed 29 references on FSW physics, tool design, defects and residual stress

Indian Institute of Space Science & Technology, Thiruvananthapuram **Aug 2025 – Present**
Teaching Assistant

- Instructing undergraduate students in CAD Lab (Semester 1) & Metrology Lab Practices (Semester 4)
- Evaluating lab-based activities and assignments of students

Tata Motors Passenger Vehicle, Chikhali, Pune **Jan 2025 – Jul 2025**
Graduate Apprenticeship Trainee — Employee No: 706714

- Served as Line In-charge at End of Line in Production Department, overseeing final quality vehicle inspection and end-of-line functional testing
- Conducted fault diagnosis and troubleshooting while maintaining constant feedback and coordination with QA/QC departments
- Implemented line balancing strategies and process improvements, ensuring quality assurance and compliance standards
- Coordinated effectively with multiple departments to optimize production workflow and resolve technical challenges

Skyline Projects, Pune **Oct 2024 – Jan 2025**
Graduate Engineer Trainee — Project In-charge

- Led project planning, preparation, and execution phases while managing material handling, verification, and safety protocols
- Supervised on-site operations ensuring quality control standards and coordinated communication across multiple teams
- Maintained comprehensive reporting and documentation while implementing technical adjustments and problem-solving strategies

CIIIT Gondawana University, Gadchiroli and GCOE, Chandrapur **Nov 2021 – Jan 2022**
Virtual Intern - CIIIT Intern

- Participated in industry-oriented training program focusing on emerging technologies and industrial applications
- IOT and Mechatronics
- Electric and Hybrid Vehicles
- Industrial Robotics

M.Tech Thesis

Scheduling of Composite Manufacturing Processes for Aerospace Components **2026 – Present**

Guides: Dr. B. S. Girish (IIST), Dr. P. Sasikumar (CMSE, ISRO) *Phase I*

- Developing a scheduling approach for autoclave and oven curing of aerospace composite parts that meets prepreg shelf-life limits and delivery deadlines while reducing energy cost
- Reviewed five recent studies and found that none combines 2D part packing, deadlines, energy use and cooling time in a single model
- Formulated a MILP that groups jobs into batches and packs them inside the autoclave to minimise makespan on a 100-job benchmark, and am implementing it in IBM CPLEX, validating on small cases first
- Defined the ISRO case with its constraints (1-month prepreg out-life, limited technicians, workstation bays, oven and autoclave fleet) and objectives (tardiness and energy cost)
- Next, building a heuristic for the benchmark problem and extending it using data from CMSE

Technical Skills

Software & Tools: MATLAB, Simulink, PLC, SCADA, SolidWorks, AutoCAD, Fusion 360, Inventor, ANSYS Fluent, ANSYS Workbench, Abaqus, MS Office Suite

Core Competencies: Manufacturing Processes, Thermodynamics, Materials Science, Quality Assurance, Process Improvement, Line Balancing, Fault Diagnosis, IoT & Mechatronics, Machine Learning, Robotics

Languages: Marathi (Native), English, Hindi

Leadership & Responsibilities

- **President, MESA (Mechanical Engineering Students' Association)** – Led student technical initiatives and organizational activities
- **Student Event Coordinator** – Organized FDP on "Autonomous Car, Connected Car & Electric Vehicle" and coordinated Blind Coding Competition

Projects

- **3D Printing of 5-Stroke 6-Cylinder Rotary IC Engine:** Design and 3D Printing
- **MeIOT Project:** Developed IoT-based mechanical system integrating sensors and actuators for real-time monitoring and control
- **Beam Engine Mechanism:** Designed and analyzed beam engine mechanism studying kinematic and dynamic characteristics

Certifications & Achievements

- Qualified GATE Mechanical Engineering 2024 (Registration No: 24S22079078, Score: 322, Marks: 26.33)

References

- **Dr. Sooraj VS** - Professor, Department of Aerospace Engineering, IIST - Trivandrum — sooraj@iist.ac.in
- **Mr. Sumeer Mehalwwal** - Deputy General Manager, Tata Motors Passenger Vehicle, Pune — mehalwwal@tatamotors.com
- **Ms. Sneha Shukla** - Shop Head, Mahindra & Mahindra — snehashukla1502@gmail.com
- **Mr. Praveen Bhuse** - Managing Director, Skyline Projects, Pune — +91 9225176363